MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, SEVEN SHIELDED TWISTED PAIRS WITH OVERALL SHIELD, NO. 16 GAUGE, FLUOROPOLYMER INSULATION, FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 7SPR-16S

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductor, tin coated, AWG size 16, class B stranding.
Second - Insulation of ethylene tetrafluoroethylene (TEFZEL). Nominal wall thickness 0.010 inch. Maximum overall diameter 0.082 inch.
Third - Two insulated conductors, one black and one white, twisted together with a nominal lay of 2-1/2 inches to form each pair.
Fourth - Braided shield of AWG no. 36 tin coated copper. Braid angle of 35 degrees minimum; coverage of 85-percent minimum.
Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.
Sixth - Seven shielded pairs cabled with a lay not greater than 24 times the pitch diameter of the layer.
Seventh - Rubber filled in order to form a firm, round assembly. Filler shall be capable of removal from the assembly without damaging the shield insulation.
Eighth - At manufacturer's option, a binder tape.
Ninth - Braided shield of AWG no. 36 tin coated copper. Braid angle of 35-degrees minimum; coverage of 85-percent minimum.

FSC 6145

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Tenth - Separator tape of polyester, applied helically with overlap.
Eleventh - Cable jacket of black polyether polyurethane. The cable jacket material shall be B.F. Goodrich Co. ESTANE 58300 or 58863; Mobay Chemical Co. TEXIN 985A. Nominal wall thickness 0.125 inch. Overall cable diameter to be 0.920 to 0.950 inch.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - at 20°C, ohms/1000 feet, maximum.	4.24
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Shield to water.....	500
Insulation dielectric (on 100 percent of insulated conductor) - test in accordance with MIL-W-81381, using a test voltage of 8.0 kV (peak), minimum.	
Insulation resistance - megohms/1000 feet, minimum.....	5000
Group A:	
Visual and dimensional.....	No failure
Group B:	
Cold bending, cable - at minus 40 ± 2°C over a 4-inch diameter mandrel.....	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum.....	5000
Elongation - percent, minimum.....	100
Cable jacket	
Tensile strength - lbf/in^2, minimum.....	5000
Elongation - percent, minimum.....	500
Group C:	
Physicals (aged)	
Insulation	
Air oven (168 hours at 80°C)	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum..	90
Cable jacket	
Air oven (168 hours at 80°C)	
This test shall be in accordance with MIL-C-915.....	No failure
Hot oil immersion	
Tensile strength, percent of unaged, minimum.....	90
Elongation, percent of unaged, minimum...	80
Shield-conformance to material coverage and construction	No failure

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QUALIFICATION INSPECTION:
Qualification inspection shall include basic electrical, plus all of groups A, B and C.

UNIT ORDERING LENGTH: 500 feet (minimum).

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-0886-08)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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